[Continuation of paper 623, "On Three-Bar Motion".]

It is to be remarked that the foregoing equation in κ determines a single position of the point K; viz. it determines $\tan\kappa$, and therefore $\sin 2\kappa$ and $\cos 2\kappa$, linearly. The equation is, in fact, a cubic equation in $\tan\kappa$, satisfied identically by $\tan\kappa = i$ and $\tan\kappa = -i$, and therefore reducible to a linear equation.

Write for a moment $\tan\kappa = \omega$, and

$$(\tan\kappa - \tan\alpha)(\tan\kappa - \tan\beta)(\tan\kappa - \tan\gamma) = \omega^3 - p\omega^2 + q\omega - r,$$
$$(\tan\kappa - \tan f)(\tan\kappa - \tan g)(\tan\kappa - \tan h) = \omega^3 - p'\omega^2 + q'\omega - r';$$

also

$$M = \cos f \cos g \cos h \;\div\; \cos\alpha\cos\beta\cos\gamma.$$

Then we have

$$\omega^3 - p\omega^2 + q\omega - r = M(\omega^3 - p'\omega^2 + q'\omega - r'),$$

where

$$p = r + M(r' - p), \qquad q = 1 + M(q' - 1).$$

Substituting these values, the equation becomes

$$\omega^3 - r\omega^2 + \omega - r = M(\omega^3 - r'\omega^2 + \omega - r'),$$

viz. dividing by $\omega^2 + 1$, this is $\omega - r = M(\omega - r')$; or substituting for r, r', M their values,

$$(\cos\alpha\cos\beta\cos\gamma - \cos f\cos g\cos h)\tan\kappa = (\sin\alpha\sin\beta\sin\gamma - \sin f\sin g\sin h),$$

which is the value of $\tan\kappa$, and then

$$\sin 2\kappa = \frac{2\tan\kappa}{1+\tan^2\kappa}, \qquad \cos 2\kappa = \frac{1-\tan^2\kappa}{1+\tan^2\kappa}.$$

It may be further noticed that, if the parabola intersect the circle in a point L, and the tangent at L to the parabola again meet the circle in M, then, if $2l$, $2m$ are the angles for the points L, M, we have l, m, m for values of f, g, h, whence l, m are determined by the equations

$$l + 2m = \alpha + \beta + \gamma, \qquad \sin(\kappa - l)\sin^2(\kappa - m) = \sin(\kappa - \alpha)\sin(\kappa - \beta)\sin(\kappa - \gamma);$$

but as the circle intersects the parabola not only in two real points, but in two other imaginary points, there is no simple formula for the determination of l and m.

To determine the linkage when the nodes are given, suppose that, in the generation by O, the vertex of the triangle OB_1C_1, we have at the node F: then, if τ, σ are the distances of C, B from the node in question, we have, as in the memoir,

$$(b_1^2 + \tau^2 - a_2^2)c_1\sigma = (c_1^2 + \sigma^2 - a_3^2)b_1\tau,$$

that is,

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b_1\tau + \sigma\tau(c_1\tau - b_1\sigma) = 0,$$

or, what is the same thing,

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b\tau + a\tau(c\tau - b\sigma) = 0.$$

Suppose, as in the figure, that F is between B and A; then, if $AF = p$, we have $c\tau = b\sigma + ap$, and the equation becomes

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b\tau + ap\sigma\tau = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if p', σ', τ' be the distances AG, BG, CG, then $b\sigma' = c\tau' + ap'$, that is, $c\tau' - b\sigma' = -ap'$, and the corresponding equation is

$$(b_1^2 - a_2^2)c\sigma' - (c_1^2 - a_3^2)b\tau' - ap'\sigma'\tau' = 0.$$

We hence find

$$(b_1^2 - a_2^2)c(\sigma\tau' - \sigma'\tau) + a\sigma\sigma'(p\tau + p'\tau') = 0,$$
$$(c_1^2 - a_3^2)b(\sigma\tau' - \sigma'\tau) + a\tau\tau'(p\sigma + p'\sigma') = 0.$$

But we have

$$BG \cdot CF = BF \cdot CG + BC \cdot FG,$$

that is,

$$\sigma'\tau = \sigma\tau' + a \cdot FG, \qquad \text{or } \sigma\tau' - \sigma'\tau = -a \cdot FG;$$

also,

$$p\sigma + p'\sigma' = AF \cdot BF + AG \cdot BG = FG \cdot BH = FG \cdot \tau'',$$

and

$$p\tau + p'\tau' = AF \cdot CF + AG \cdot CG = FG \cdot CH = FG \cdot \sigma'',$$

as may be shown without difficulty, p'', σ'', τ'' being the distances AH, BH, CH. Hence the equations become

$$c(b_1^2 - a_2^2) - \sigma\sigma'\sigma'' = 0,$$
$$b(c_1^2 - a_3^2) - \tau\tau'\tau'' = 0,$$

showing that, the foci being as in the figure, $b_1^2 - a_2^2$ and $c_1^2 - a_3^2$ are each of them positive; viz. that, in the generation by the triangle OC_1B_1, the radial bars a_2, a_3 are shorter than the sides b_1, c_1 respectively. Substituting for b_1, &c. the values $k_1 \sin B$, &c.; also, instead of k_1, k_2, k_3, introducing the quantities λ_1, λ_2, λ_3, where

$$k_1, k_2, k_3 = \lambda_1 \sin A,\ \lambda_2 \sin B,\ \lambda_3 \sin C,$$

these equations become

$$c(\lambda_1^2 - \lambda_2^2) \sin^2 A \sin^2 B = \sigma\sigma'\sigma'',$$
$$b(\lambda_1^2 - \lambda_3^2) \sin^2 A \sin^2 C = \tau\tau'\tau'';$$

or, as these may be written, putting for shortness $M = \sin A \sin B \sin C$,

$$M^2k^2(\lambda_1^2 - \lambda_2^2) = c\,\sigma\sigma'\sigma'',$$
$$M^2k^2(\lambda_1^2 - \lambda_3^2) = b\,\tau\tau'\tau''.$$

All the quantities have so far been regarded as positive, and the formulae are applicable to the particular figure; but, to present them in a form applicable to any order of the nodes and foci, we have only to write the equations in the forms

$$M^2(\lambda_1^2 - \lambda_2^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$
$$M^2(\lambda_1^2 - \lambda_3^2) = k^2 \sin(\alpha - \gamma) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta).$$

And these may be replaced by the system

$$M^2(\lambda_2^2 - \lambda_3^2) = k^2 \sin(\beta - \gamma) \sin(f - \alpha) \sin(g - \alpha) \sin(h - \alpha),$$
$$M^2(\lambda_3^2 - \lambda_1^2) = k^2 \sin(\gamma - \alpha) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta),$$
$$M^2(\lambda_1^2 - \lambda_2^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$M^2k^2(\lambda_2^2 - \lambda_3^2) = BC \cdot FA \cdot GA \cdot HA,$$
$$M^2k^2(\lambda_3^2 - \lambda_1^2) = CA \cdot FB \cdot GB \cdot HB,$$
$$M^2k^2(\lambda_1^2 - \lambda_2^2) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they stand for $k\sin(\beta-\gamma)$, $k\sin(f-\alpha)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$\begin{aligned} BC\,(c_1^2-b_1^2) &= FA\cdot GA\cdot HA,\\ CA\,(a_3^2-c_1^2) &= FB\cdot GB\cdot HB,\\ AB\,(b_1^2-a_2^2) &= FC\cdot GC\cdot HC,\end{aligned}$$

where the signs are as just mentioned. We may say that $\pm(a_2^2-b_1^2)$ is the modulus for the focus A; and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624.

ON THE BICURSAL SEXTIC.

[From the Proceedings of the London Mathematical Society, vol. vii. (1875–1876), pp. 166–172. Read March 10, 1876.]

In the paper "On the mechanical description of certain sextic curves," *Proceedings of the London Mathematical Society*, vol. iv. (1872), pp. 105–111, [504], I obtained the bicursal sextic as a rational transformation of a binodal quartic. The theory was in effect as follows: taking P, Q, R, each of them a function of λ, μ of the form $(*\quad)(\lambda,1)^2(1,\mu)^2$, and considering (λ,μ) as connected by the equation $\Omega=0$, (viz. λ, μ being coordinates, this represents a binodal quartic), then, if we assume $x:y:z=P:Q:R$, the locus of the point (x,y,z) is a curve rationally connected with the binodal quartic, viz. the points of the two curves have with each other a $(1,1)$ correspondence; whence the locus in question, say the curve $U=0$, is bicursal. The degree is obtained as the number of the intersections of the curve by an arbitrary line, or, what is the same thing, the number of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega=0,\qquad \alpha P+\beta Q+\gamma R=0,$$

viz. each of these being a quartic curve having the same two nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4\cdot4-2\cdot4=8$, and thus the curve $U=0$ is in general of the order 8. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the nodes) k common intersections, then these are also fixed intersections of the two curves $\Omega=0$, $\alpha P+\beta Q+\gamma R=0$, and the number of variable intersections is reduced to $8-k$: we have thus $8-k$ as the order of the curve $U=0$. In particular, if $k=2$, then the curve is a bicursal sextic.

The theory assumes a different and more simple form if, in the several functions Ω, P, Q, R, we suppose that the terms in λ^2, μ^2 are wanting. The curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ are here cubics having two common points; the curve $U=0$, qua rational transformation of the cubic $\Omega=0$, is still a bicursal curve; but its order is given as the number of the variable intersections of the cubics

$$\Omega=0,\qquad \alpha P+\beta Q+\gamma R=0,$$

viz. this is $=3\cdot3-2=7$. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $=7-k$: and this is therefore the order of the curve $U=0$. In particular, if $k=1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega=0$, $x:y:z=P:Q:R$, where $\Omega=0$, $P=0$, $Q=0$, $R=0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $\alpha P+\beta Q+\gamma R=0$, there are any two curves which have 3 common intersections with the curve $\Omega=0$. (Observe that we throughout disregard the 3 common points of the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$, and attend only to the 6 variable points of intersection of the curves $\Omega = 0$ and $\alpha P + \beta Q + \gamma R = 0$,—the meaning is, that there are two curves of the series such that, attending only to the 6 variable intersections of each of them with the curve $\Omega = 0$, there are three common intersections.) For, supposing the two curves to be $\alpha P + \beta Q + \gamma R = 0$ and $\alpha' P + \beta' Q + \gamma' R = 0$, then any curve whatever

$$\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$$

has the same three intersections with the curve $\Omega = 0$, say these are the points A_1, A_2, A_3, the coordinates of which are independent of θ. Hence the line

$$(\alpha x + \beta y + \gamma z) + \theta(\alpha' x + \beta' y + \gamma' z) = 0$$

intersects the curve $U = 0$ in six points, three of which, as corresponding to the points A_1, A_2, A_3, are independent of θ, viz. they are the same three points for any line whatever of the series; and this means that the curve $U = 0$ has at the point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0)$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $\alpha x + \beta y + \gamma z + \theta(\alpha' x + \beta' y + \gamma' z) = 0$, rationally determine θ so that one of the three variable points of intersection shall correspond to A_1, A_2, or A_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points A_1, A_2, A_3. The three lines thus determined are the three tangents to the curve at the triple point: and the three branches may be considered as corresponding to the three points A_1, A_2, A_3, respectively.

There is no loss of generality in assuming that the triple point is the point $(x = 0,\ z = 0)$; the condition then simply is that the curves $P = 0$, $R = 0$ shall have three common intersections with the curve $\Omega = 0$; and the tangents at the triple point are $x + \theta z = 0$, θ being so determined that one of the three variable points of intersection shall correspond to one of the three points A_1, A_2, A_3: in particular, if this is the case for the line $x = 0$, then this line will be one of the tangents at the triple point.

The bicursal sextic may have a second triple point, viz. three other nodes may unite together into a triple point. The theory is precisely the same: we must have two other curves $\alpha P + \beta Q + \gamma R = 0$, $\alpha' P + \beta' Q + \gamma' R = 0$, having with the curve $\Omega = 0$ three common intersections B_1, B_2, B_3: there is then a second triple point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0)$$

and, to find the tangents at this point, we must determine θ so that one of the variable points of intersection of the line

$$\alpha x + \beta y + \gamma z + \theta(\alpha' x + \beta' y + \gamma' z) = 0$$

with the sextic shall correspond with B_1, B_2, or B_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points B_1, B_2, B_3. In particular, if, as before, the curves $P = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, and if, moreover, the curves $Q = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, then the bicursal sextic will have the two triple points $(x = 0,\ z = 0)$ and $(y = 0,\ z = 0)$; and it may further happen that the line $x = 0$ is a tangent at the first triple point, and the line $y = 0$ a tangent at the second triple point. The sextic may in like manner have a third triple point, but this is a special case which I do not at present consider.

I write for greater convenience $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, so as to make Ω, P, Q, R each of them a homogeneous cubic function of (λ, μ, ν); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion; viz. the equations $\Omega = 0$, $x : y : z = P : Q : R$ are respectively taken to be

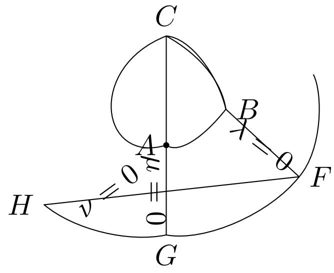
$$\nu(h\nu\lambda + f\lambda^2) + \mu(d\lambda\mu + e\nu\lambda + g\lambda^2) + \mu^2(f\nu + h\lambda) = 0,$$

$$x : y : z = \lambda\mu(a\lambda + b\mu) : \nu^2(c\lambda + d\mu) : \lambda\mu\nu.$$

The four curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have thus the three common intersections

$$(\mu = 0,\ \nu = 0), \quad (\nu = 0,\ \lambda = 0), \quad (\lambda = 0,\ \mu = 0),$$

represented in the figure by the points A, B, C; the curve drawn in the figure is the curve $\Omega = 0$, and the points F, G, H are the third points of intersection of the cubic with the lines BC, CA, AB respectively.



The equation $P+\theta R=0$ is here $\lambda\mu(a\lambda+b\mu+\theta\nu)=0$, which intersects $\Omega=0$ in the points C, A, G, C, B, F, and the three intersections by the line $a\lambda+b\mu+\theta\nu=0$; viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and the three intersections by the line. Hence, of the six intersections, we have C, F, G independent of θ, or we have ($x=0$, $z=0$) a triple point, say I, corresponding to the three points C, F, G, viz. these are the points

$$(\lambda,\mu,\nu)=(0,0,1),\ (0,g,-f),\ (h,0,-f),\qquad (C,F,G).$$

The equation $Q+\theta R=0$ is $\nu(c\lambda\nu+d\mu\nu+\theta\lambda\mu)=0$; viz. the line $\nu=0$ meets $\Omega=0$ in the three points A, B, H, and the conic $c\lambda\nu+d\mu\nu+\theta\lambda\mu=0$ meets $\Omega=0$ in the points A, B, C, and three other points: hence, rejecting the points A, B, C, the six points of intersection are the points A, B, H, and the three variable points of intersection by the conic; or we have ($y=0$, $z=0$) a triple point, say J, corresponding to the three points A, B, H, viz. these are the points

$$(\lambda,\mu,\nu)=(1,0,0),\ (0,1,0),\ (h,-g,0),\qquad (A,B,H).$$

To find the tangents at the triple point I, these are $x+\theta z=0$, where θ is to be successively determined by the conditions that the line $a\lambda+b\mu+\theta\nu=0$ shall pass through the points C, F, G[1]; viz. we thus have

$$\begin{array}{lll}
\theta=0, & x=0, & \text{tangent corresponding to the point } C,(0,0,1),\\
\theta=\dfrac{a}{f}, & fx+bgz=0, & \text{tangent corresponding to the point } F,(0,g,-f),\\
\theta=\dfrac{a}{f}, & fx+ahz=0, & \text{tangent corresponding to the point } G,(h,0,-f).
\end{array}$$

And similarly, at the triple point J, the tangents are $y+\theta z=0$, where θ is to be successively determined by the conditions that the conic $c\nu\lambda+d\nu\mu+\theta\lambda\mu=0$ shall pass through the point H, and shall touch the cubic at the points A, B; viz. we thus have

$$\begin{array}{lll}
\theta=0, & y=0, & \text{tangent corresponding to the point } H,(h,-g,0),\\
\theta=\dfrac{c}{f}, & fy+cgz=0, & \text{tangent corresponding to the point } A,(1,0,0),\\
\theta=\dfrac{d}{f}, & fy+dhz=0, & \text{tangent corresponding to the point } B,(0,1,0).
\end{array}$$

The two last values of θ are obtained by the consideration that the equations of the tangents to $\Omega=0$ at the points A, B respectively, are $g\mu+f\nu=0$, $h\lambda+f\nu=0$, where λ, μ, ν are current coordinates of a point on the tangent: it may be added that the equation of the tangent at the point C is $h\lambda+g\mu=0$.

The three-bar curve may be represented by means of a system of equations of the last-mentioned form, viz. $x:y:z=\lambda\mu(a\lambda+b\mu):\nu^2(c\lambda+d\mu):\lambda\mu\nu$, where λ, μ, ν are connected as above; or, taking X, Y as ordinary rectangular coordinates, x, y, and z are here the circular coordinates $\dfrac{x}{z}=X+iY$, $\dfrac{y}{z}=X-iY$, and $z=1$; and the parameters $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ denote like functions $\cos\theta+i\sin\theta$, $\cos\phi+i\sin\phi$ of angles θ, ϕ which

[1] Observe the somewhat altered form of the condition: θ is to be determined so that the cubic $\lambda\mu(a\lambda+b\mu+\theta\nu)=0$ shall touch the cubic $\Omega=0$ at one of the points C, F, G: but, as the first-mentioned cubic breaks up, and the component curve $a\lambda+b\mu+\theta\nu=0$ does not pass through any one of these points, this can only mean that θ shall be so determined as that the line shall pass through one of these points, viz. that there shall be at the point, not a proper contact, but a double intersection, arising from a node of the cubic $\lambda\mu(a\lambda+b\mu+\theta\nu)=0$. And the like case happens for the other triple point; viz. there the cubic $\nu(c\nu\lambda+d\nu\mu+\theta\lambda\mu)=0$ is to touch the cubic $\Omega=0$ at one of the points A, B, H; the component conic $c\nu\lambda+d\nu\mu+\theta\lambda\mu=0$ passes through the points A and B but not through H; hence the conditions for θ are, that the conic shall touch the cubic at A or B, or that it shall pass through H.

are the inclinations of two bars to a fixed line. Using, for convenience, Figure 2 of my paper on Three-bar Motion, (p. 553 of this volume), the curve is considered as the locus of the vertex O of the triangle OC_1B_1, connected by the bars C_1C and B_1B with the fixed points B and C respectively; and we have $CC_1 = a_2$, $OC_1 = b_1$, $OB_1 = c_1$, $C_1B_1 = a_1$, $B_1B = a_3$. Also, to avoid confusion with the foregoing notation of the present paper, instead of calling it a, I take $BC = a_0$: the angle OC_1B_1 is $= C_1$, and $\cos C_1 + i\sin C_1$ is taken $= \gamma_1$.

Hence, taking the origin at C, the axis of X coinciding with CB and that of Y being at right angles to it: taking also θ, ϕ, ψ for the inclinations of CC_1, C_1B_1, and B_1B to CB, we have

$$a_2\cos\theta + a_1\cos\phi - a_0 = -a_3\cos\psi,$$
$$a_2\sin\theta + a_1\sin\phi = a_3\sin\psi;$$

viz. writing $\cos\theta + i\sin\theta = \lambda$, $\cos\phi + i\sin\phi = \mu$, these give

$$a_2\lambda + a_1\mu - a_0 = -a_3(\cos\psi - i\sin\psi),$$
$$a_2\frac{1}{\lambda} + a_1\frac{1}{\mu} - a_0 = -a_3(\cos\psi + i\sin\psi),$$

that is,

$$(a_2\lambda + a_1\mu - a_0)\left(a_2\tfrac{1}{\lambda} + a_1\tfrac{1}{\mu} - a_0\right) - a_3^2 = 0;$$

viz.

$$(a_2^2 + a_1^2 + a_0^2 - a_3^2) + a_1a_2\left(\tfrac{\lambda}{\mu} + \tfrac{\mu}{\lambda}\right) - a_0a_2\left(\lambda + \tfrac{1}{\lambda}\right) - a_0a_1\left(\mu + \tfrac{1}{\mu}\right) = 0,$$

for the relation between the parameters λ, μ. And then

$$X = a_2\cos\theta + b_1\cos(\phi + C_1),$$
$$Y = a_2\sin\theta + b_1\sin(\phi + C_1);$$

viz. if $x, y = X + iY,\ X - iY$, then

$$x = a_2\lambda + b_1\gamma_1\mu, \qquad y = a_2\tfrac{1}{\lambda} + b_1\frac{1}{\gamma_1\mu},$$

which equations determine the coordinates (x, y) in terms of the parameters λ, μ connected by the foregoing relation.

Writing for homogeneity $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, and $\dfrac{x}{z}$, $\dfrac{y}{z}$ in place of x, y, the equations become

$$(a_2^2 + a_1^2 + a_0^2 - a_3^2)\lambda\mu\nu + a_1a_2(\lambda^2 + \mu^2)\nu - a_0a_1\mu(\nu^2 + \lambda^2) - a_0a_2\lambda(\mu^2 + \nu^2) = 0,$$

and

$$x : y : z = (a_2\lambda + b_1\gamma_1\mu)\lambda\mu : \left(\tfrac{a_2}{\gamma_1}\lambda + a_1\mu\right)\nu^2 : \lambda\mu\nu.$$

Comparing with the foregoing equations

$$e\lambda\mu\nu + f(\lambda^2 + \mu^2)\nu + g(\nu^2 + \lambda^2)\mu + h(\mu^2 + \nu^2)\lambda = 0,$$

and

$$x : y : z = (a\lambda + b\mu)\lambda\mu : (c\lambda + d\mu)\nu^2 : \lambda\mu\nu,$$

the equations agree together, and we have

$$\begin{aligned}
e &= a_2^2 + a_1^2 + a_0^2 - a_3^2,\\
f &= +a_1a_2,\\
g &= -a_0a_2,\\
h &= -a_0a_1,\\
a &= a_2,\\
b &= b_1\gamma_1,\\
c &= \frac{a_2}{\gamma_1},\\
d &= a_1.
\end{aligned}$$

The tangents at the triple points thus are

$$\begin{aligned}&x = 0,\\ &a_1 x - a_0 b_1 \gamma_1 z = 0, \qquad a_1 y - \frac{a_0 a_1}{\gamma_1} z = 0,\\ &x - a_0 z = 0, \qquad y - a_0 z = 0;\end{aligned}$$

viz. restoring the rectangular coordinates, and for γ substituting the value $\cos C + i \sin C$, for a_0 writing a, and taking $b = \dfrac{b_1\gamma_1}{\gamma}$, we have

$$\begin{aligned}&X + iY = 0, && X - iY = 0,\\ &X + iY = b(\cos C + i\sin C), && X - iY = b(\cos C - i\sin C),\\ &X + iY = a, && X - iY = a;\end{aligned}$$

viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b\cos C, b\sin C)$, the third two in the point $(a, 0)$: the first and third of these are the points B and C, the second of them is the point A of the figure: viz. the formulae give the point A, forming, with B and C, a triad of foci.

625.

On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines.

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In a congruency or doubly infinite system of right lines, the direction-cosines α, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $\alpha\, dx + \beta\, dy + \gamma\, dz$ must be an exact differential: when this is so, writing $V = \int(\alpha\, dx + \beta\, dy + \gamma\, dz)$, we have $V = c$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface exists, its differential equation is $\alpha\, dx + \beta\, dy + \gamma\, dz = 0$, and this equation must therefore be integrable by a factor. Now the functions α, β, γ are such that $\alpha^2 + \beta^2 + \gamma^2 = 1$, and they besides satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. passing from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x + \rho\alpha$, $y + \rho\beta$, $z + \rho\gamma$ (ρ infinitesimal), the line belonging to this point is the original line; and consequently α, β, γ, considered as functions of x, y, z, must remain unaltered when these variables are changed into $x + \rho\alpha$, $y + \rho\beta$, $z + \rho\gamma$, respectively. We thus obtain the equations

$$\begin{aligned}\alpha\frac{d\alpha}{dx} + \beta\frac{d\alpha}{dy} + \gamma\frac{d\alpha}{dz} &= 0,\\ \alpha\frac{d\beta}{dx} + \beta\frac{d\beta}{dy} + \gamma\frac{d\beta}{dz} &= 0,\\ \alpha\frac{d\gamma}{dx} + \beta\frac{d\gamma}{dy} + \gamma\frac{d\gamma}{dz} &= 0.\end{aligned}$$

Combining herewith the equations obtained by differentiation of $\alpha^2 + \beta^2 + \gamma^2 = 1$, viz.

$$\begin{aligned}\alpha\frac{d\alpha}{dx} + \beta\frac{d\beta}{dx} + \gamma\frac{d\gamma}{dx} &= 0,\\ \alpha\frac{d\alpha}{dy} + \beta\frac{d\beta}{dy} + \gamma\frac{d\gamma}{dy} &= 0,\\ \alpha\frac{d\alpha}{dz} + \beta\frac{d\beta}{dz} + \gamma\frac{d\gamma}{dz} &= 0,\end{aligned}$$

and subtracting the corresponding equations, we obtain three equations which may be written

$$\beta\left(\frac{d\beta}{dz}-\frac{d\gamma}{dy}\right)=\gamma\left(\frac{d\gamma}{dx}-\frac{d\alpha}{dz}\right)=\alpha\left(\frac{d\alpha}{dy}-\frac{d\beta}{dx}\right),$$

or, what is the same thing,

$$\frac{d\beta}{dz}-\frac{d\gamma}{dy}=k\alpha,\quad \frac{d\gamma}{dx}-\frac{d\alpha}{dz}=k\beta,\quad \frac{d\alpha}{dy}-\frac{d\beta}{dx}=k\gamma,$$

and, multiplying by α, β, γ, and adding,

$$k=\alpha\left(\frac{d\beta}{dz}-\frac{d\gamma}{dy}\right)+\beta\left(\frac{d\gamma}{dx}-\frac{d\alpha}{dz}\right)+\gamma\left(\frac{d\alpha}{dy}-\frac{d\beta}{dx}\right).$$

We thus see that, if the function on the right-hand vanish, then $k=0$, and consequently also

$$\frac{d\beta}{dz}-\frac{d\gamma}{dy}=0,\quad \frac{d\gamma}{dx}-\frac{d\alpha}{dz}=0,\quad \frac{d\alpha}{dy}-\frac{d\beta}{dx}=0,$$

viz. if the equation $\alpha\,dx+\beta\,dy+\gamma\,dz=0$ be integrable, then $\alpha\,dx+\beta\,dy+\gamma\,dz$ is an exact differential; which is the theorem in question.

But it is interesting to obtain the first mentioned set of differential equations from the analytical equations of a congruency, viz. these are $x=mz+p$, $y=nz+q$, where m, n, p, q are functions of two arbitrary parameters, or, what is the same thing, p, q are given functions of m, n; and therefore, from the three equations, m, n are given functions of x, y, z. And it is also interesting to express in terms of these quantities m, n, considered as functions of x, y, z, the condition for the existence of the set of surfaces.

We have

$$\frac{\alpha}{m}=\frac{\beta}{n}=\frac{\gamma}{1},$$

and thence without difficulty

$$\alpha,\,\beta,\,\gamma=\frac{m}{R},\ \frac{n}{R},\ \frac{1}{R},\qquad \text{where } R=\sqrt{1+m^2+n^2};$$

$$\left(\alpha\frac{d}{dx}+\beta\frac{d}{dy}+\gamma\frac{d}{dz}\right)\frac{1}{R}=-\frac{1}{R^3}\left[m\left(m\frac{dm}{dx}+n\frac{dm}{dy}+\frac{dm}{dz}\right)+n\left(m\frac{dn}{dx}+n\frac{dn}{dy}+\frac{dn}{dz}\right)\right],$$

so that the required equations in α, β, γ will be satisfied if only

$$m\frac{dm}{dx}+n\frac{dm}{dy}+\frac{dm}{dz}=0,$$

$$m\frac{dn}{dx}+n\frac{dn}{dy}+\frac{dn}{dz}=0,$$

and it is to be shown that these equations hold good.

Writing for shortness $dp=A\,dm+B\,dn$, $dq=C\,dm+D\,dn$, the equations of the line give

$$\begin{aligned}
0&=z\frac{dm}{dx}+A\frac{dm}{dx}+B\frac{dn}{dx}, & 0&=z\frac{dn}{dx}+C\frac{dm}{dx}+D\frac{dn}{dx},\\
0&=z\frac{dm}{dy}+A\frac{dm}{dy}+B\frac{dn}{dy}, & 0&=z\frac{dn}{dy}+C\frac{dm}{dy}+D\frac{dn}{dy},\\
-m&=z\frac{dm}{dz}+A\frac{dm}{dz}+B\frac{dn}{dz}, & -n&=z\frac{dn}{dz}+C\frac{dm}{dz}+D\frac{dn}{dz};
\end{aligned}$$

or, writing

$$\lambda,\mu,\nu=\frac{dm}{dy}\frac{dn}{dz}-\frac{dm}{dz}\frac{dn}{dy},\ \frac{dm}{dz}\frac{dn}{dx}-\frac{dm}{dx}\frac{dn}{dz},\ \frac{dm}{dx}\frac{dn}{dy}-\frac{dm}{dy}\frac{dn}{dx},$$

so that identically

$$\lambda\frac{dm}{dx}+\mu\frac{dm}{dy}+\nu\frac{dm}{dz}=0,$$

$$\lambda\,\frac{dn}{dx}+\mu\,\frac{dn}{dy}+\nu\,\frac{dn}{dz}=0,$$

then in each set, multiplying by λ, μ, ν and adding, so as to eliminate A, B, C, D, we find

$$\lambda-m\nu=0,\qquad \mu-n\nu=0.$$

Substituting these values of λ, μ in the last preceding equations, ν divides out, and we have the two equations in question.

The foregoing equations give further

$$A,B,C,D=-z+\frac{1}{\nu}\frac{dn}{dy},\ -\frac{1}{\nu}\frac{dm}{dy},\ -\frac{1}{\nu}\frac{dm}{dx},\ -z+\frac{1}{\nu}\frac{dm}{dx}.$$

Taking for α, β, γ the before-mentioned values, we find

$$\begin{aligned}\frac{d\alpha}{dy}-\frac{d\beta}{dx}&=\frac{1}{R}\left(\frac{dm}{dy}-\frac{dn}{dx}\right)-\frac{m}{R^3}\,r\left(\frac{dm}{dy}-\frac{dn}{dy}\right)-\frac{n}{R^3}\,r'\left(\frac{dm}{dx}-\frac{dn}{dx}\right)\\&=\frac{1}{R^3}\left\{(\,\cdot\,)\,\frac{dm}{dy}+(\,\cdot\,)\frac{dn}{dx}+\left(\frac{dm}{dx}-\frac{dn}{dy}\right)\right\};\end{aligned}$$

and similarly, but using the equations

$$m\,\frac{dm}{dx}+n\,\frac{dm}{dy}+\frac{dm}{dz}=0,\qquad m\,\frac{dn}{dx}+n\,\frac{dn}{dy}+\frac{dn}{dz}=0$$

to eliminate the coefficients $\dfrac{dm}{dz}$, $\dfrac{dn}{dz}$ which in the first instance present themselves, we find

$$\frac{d\beta}{dz}-\frac{d\gamma}{dy}=\frac{n}{R^3}(\,\cdot\,),\qquad \frac{d\gamma}{dx}-\frac{d\alpha}{dz}=-\frac{1}{R^3}(\,\cdot\,);$$

whence, multiplying by γ, α, β, and adding,

$$\begin{aligned}&\alpha\left(\frac{d\beta}{dz}-\frac{d\gamma}{dy}\right)+\beta\left(\frac{d\gamma}{dx}-\frac{d\alpha}{dz}\right)+\gamma\left(\frac{d\alpha}{dy}-\frac{d\beta}{dx}\right)\\&=\frac{1}{1+m^2+n^2}\left\{(\,\cdot\,)\frac{dm}{dy}+(\,\cdot\,)\frac{dn}{dx}+\left(\frac{dm}{dx}-\frac{dn}{dy}\right)\right\};\end{aligned}$$

or we have

$$(1+n^2)\frac{dm}{dy}-(1+m^2)\frac{dn}{dx}+mn\left(\frac{dm}{dx}-\frac{dn}{dy}\right)=0$$

as the condition for the existence of the set of surfaces.

It is clear that the condition is satisfied when the lines are the normals of a given surface: seeking the surfaces which cut the lines at right angles, we obtain the parallel surfaces; and we are led to the theorem that any parallel surface is the locus of the extremity of a line of constant length measured off from each point of the surface along the normal—or, what is equivalent thereto, the parallel surface is the envelope of a sphere of constant radius having its centre on the surface. I will verify the theorem for the case of the ellipsoid. Taking X, Y, Z as the coordinates of a point on the ellipsoid $\dfrac{X^2}{a^2}+\dfrac{Y^2}{b^2}+\dfrac{Z^2}{c^2}=1$, and x, y, z as current coordinates, the equations of the normal are

$$\frac{a^2}{X}(x-X)=\frac{b^2}{Y}(y-Y)=\frac{c^2}{Z}(z-Z),\quad (=\lambda,\ \text{suppose}).$$

We have therefore

$$X,Y,Z=\frac{a^2x}{a^2+\lambda},\ \frac{b^2y}{b^2+\lambda},\ \frac{c^2z}{c^2+\lambda},$$

and thence

$$\frac{a^2x^2}{(a^2+\lambda)^2}+\frac{b^2y^2}{(b^2+\lambda)^2}+\frac{c^2z^2}{(c^2+\lambda)^2}=1,$$

an equation which determines λ as a function of x, y, z.

The direction-cosines α, β, γ of the normal are proportional to $\frac{X}{a^2}, \frac{Y}{b^2}, \frac{Z}{c^2}$, that is, to $\frac{x}{a^2+\lambda}$, $\frac{y}{b^2+\lambda}$, $\frac{z}{c^2+\lambda}$, and the equation $\alpha^2+\beta^2+\gamma^2=1$ then determines their absolute magnitudes: the equation $\alpha\,dx+\beta\,dy+\gamma\,dz=dV$ thus is

$$\frac{\dfrac{x\,dx}{a^2+\lambda}+\dfrac{y\,dy}{b^2+\lambda}+\dfrac{z\,dz}{c^2+\lambda}}{\sqrt{\dfrac{x^2}{(a^2+\lambda)^2}+\dfrac{y^2}{(b^2+\lambda)^2}+\dfrac{z^2}{(c^2+\lambda)^2}}}=dV,$$

viz. the left-hand side, considering therein λ as a given function of V, is an exact differential. We verify this by finding the value of V, viz. writing down the two equations

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}+\frac{z^2}{c^2+\lambda}-\frac{V^2}{\lambda}=0,$$

$$\frac{x^2}{(a^2+\lambda)^2}+\frac{y^2}{(b^2+\lambda)^2}+\frac{z^2}{(c^2+\lambda)^2}-\frac{V^2}{\lambda^2}=1,$$

these are equivalent in virtue of the equation that determines λ: and it is to be shown that, regarding V as given by either of them, say by the second equation, we have for dV its foregoing value. In fact, differentiating the second equation, the term in $d\lambda$ disappears by virtue of the first equation, and the result is

$$\frac{x\,dx}{a^2+\lambda}+\frac{y\,dy}{b^2+\lambda}+\frac{z\,dz}{c^2+\lambda}-\frac{V\,dV}{\lambda}=0,$$

in which substituting for $\frac{V}{\lambda}$ its value from the first equation, we have for dV the value in question. Regarding V as a given constant, the two equations give, by elimination of λ, an equation $\phi(x,y,z,V)=0$, which is, in fact, the surface parallel to the ellipsoid and at a constant normal distance $=V$ from it.

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On the General Differential Equation $\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0$, Where X, Y are the Same Quartic Functions of x, y Respectively.

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Write $\Theta=a+b\theta+c\theta^2+d\theta^3+e\theta^4$, the general quartic function of θ; and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0.$$

We have

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ w^2, & w, & 1, & \sqrt{W} \end{vmatrix}=0,$$

a particular integral of

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dz}{\sqrt{Z}}+\frac{dw}{\sqrt{W}}=0;$$

and consequently the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0,$$

viz. the two constants z, w must enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

Take $w = \infty$; the equation becomes

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{e} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$e(\theta^2 + \beta\theta + \gamma)^2 - (e\theta^4 + d\theta^3 + c\theta^2 + b\theta + a) = (2\sqrt{e}\,\beta - d)(\theta - x)(\theta - y)(\theta - z),$$

or, what is the same thing,

$$\begin{aligned} e\,(2\gamma + \beta^2) - c &= -(2\beta\sqrt{e} - d)(x + y + z), \\ e\,\cdot 2\beta\gamma - b &= (2\beta\sqrt{e} - d)(yz + zx + xy), \\ e\gamma^2 - a &= -(2\beta\sqrt{e} - d)\,xyz, \end{aligned}$$

where β, γ are indeterminate coefficients which are to be eliminated.

Write

$$x^2 = \frac{\sqrt{X}}{\sqrt{e}} = P, \qquad y^2 = \frac{\sqrt{Y}}{\sqrt{e}} = Q;$$

then we have

$$\beta x + \gamma + P = 0, \qquad \beta y + \gamma + Q = 0;$$

giving

$$\beta : \gamma : 1 = Q - P : Py - Qx : x - y.$$

Substituting these values in the first of the preceding three equations, we have

$$e\,\frac{2(Py - Qx)(x - y) + (Q - P)^2}{(x - y)^2} - c = -\left\{\frac{2(Q - P)\sqrt{e}}{x - y} - d\right\}(x + y + z),$$

that is,

$$e\left\{\frac{2(Qy - Px)}{x - y} + \frac{(Q - P)^2}{(x - y)^2}\right\} - c = -\left\{\frac{2(Q - P)\sqrt{e}}{x - y} - d\right\}(x + y + z);$$

or, reducing by

$$Qy - Px = y^2 - x^2 + \frac{x\sqrt{X} - y\sqrt{Y}}{\sqrt{e}},$$

$$Q - P = \frac{\sqrt{X} - \sqrt{Y}}{\sqrt{e}}, \qquad = \frac{x^2 - y^2 + (\cdots)}{\sqrt{e}\,(x - y)} \quad \text{if } M = \frac{\sqrt{X} - \sqrt{Y}}{x - y},$$

this is

$$e\left\{\frac{2(x\sqrt{X} - y\sqrt{Y})}{\sqrt{e}\,(x - y)} - \frac{x + y}{e}\right\}^2 + 2xy + \frac{1}{e}M^2 - \frac{2(x + y)}{\sqrt{e}}M\sqrt{X} + \frac{2}{\sqrt{e}}M\sqrt{Y} = c + d(x + y + z) + e(x + y)^2.$$

We have Euler's solution in the far more simple form

$$M^2 = C + d(x + y) + e(x + y)^2,$$

where C is the arbitrary constant. It is to be observed that, in the particular case where $e = 0$, the first equation becomes

$$M^2 = c + d(x + y + z);$$

and the two results for this case agree on putting $C = c + dz$.

But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, in my *Treatise on Elliptic Functions*, Chap. xiv., further developed the theory of Euler's solution, and have shown that, regarding C as variable, and writing

$$\mathfrak{S} = ad^2 + b^2e - 2bcd + C[-4ae + bd + (C-c)^2],$$

then the given equation between the variables x, y, C corresponds to the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\mathfrak{S}}} = 0,$$

a result which will be useful for effecting the identification. The Abelian solution may be written

$$e\frac{2(x\sqrt{X} - y\sqrt{Y})}{x-y} - e\left\{2xy - x^2 - y^2 + \frac{1}{e}M^2 - 2(x+y)\frac{1}{\sqrt{e}}M\right\} \\ - c - d(x+y) = z\big[d + 2e(x+y) - 2M\sqrt{e}\big]\,;$$

and substituting for M its value, and multiplying by $(x-y)^2$, the equation becomes

$$2\sqrt{e}\,(x-y)(x\sqrt{X} - y\sqrt{Y}) - e(x^2+y^2)(x-y)^2 + (\sqrt{X} - \sqrt{Y})^2 \\ - 2(x^2-y^2)(\sqrt{X}-\sqrt{Y})\sqrt{e} - c(x-y)^2 - d(x+y)(x-y)^2 \\ = z(x-y)\Big[d(x-y) + 2e(x^2-y^2) - 2(\sqrt{X}-\sqrt{Y})\sqrt{e}\Big]\,.$$

On the left-hand side, the rational part is

$$X + Y + c(-x^2 + 2xy - y^2) + d(-x^3 + x^2y + xy^2 - y^3) + e(-x^4 + 2x^3y - 2x^2y^2 + 2xy^3 - y^4)$$

which, substituting therein for X, Y their values, becomes

$$= 2a + b(x+y) + c\cdot 2xy + dxy(x+y) + e\cdot 2xy(x^2 - xy + y^2);$$

and the irrational part is at once found to be

$$= 2\sqrt{e}\,(x-y)(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{XY}.$$

The equation thus is

$$2a + b(x+y) + c\cdot 2xy + dxy(x+y) + e\cdot 2xy(x^2 - xy + y^2) \\ + 2\sqrt{e}\,(x-y)(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{XY},$$

$$z = \frac{\text{(above expression)}}{(x-y)\Big[d(x-y) + 2e(x^2-y^2) - 2(\sqrt{X}-\sqrt{Y})\sqrt{e}\Big]},$$

which equation is thus a form of the general integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, and the particular integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0.$$

Multiplying the numerator and the denominator by

$$d(x-y) + 2e(x^2-y^2) + 2(\sqrt{X}-\sqrt{Y})\sqrt{e},$$

the denominator becomes

$$= (x-y)^2\left[\{d + 2e(x+y)\}^2 - 4e\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2\right]$$

which, introducing herein the C of Euler's equation, is

$$= (x-y)^2(d^2-4eC).$$

We have therefore

$$\begin{aligned} z(x-y)^2(d^2-4eC) = &\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2) \\ &+2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})-2\sqrt{XY}\Big\} \\ &\times\Big\{d(x-y)+2e(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\Big\}. \end{aligned}$$

Using $\mathfrak{S}$ to denote the same value as before, the function on the right-hand is, in fact,

$$= (x-y)^2\Big\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\Big\},$$

and, this being so, the required relation between z, C is

$$z(d^2-4eC) = \Big\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\Big\}.$$

To prove this, we have first, from the equation

$$M^2 = C+d(x+y)+e(x+y)^2,$$

to express C as a function of x, y. This equation, regarding therein C as a variable, gives

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dC}{\sqrt{\mathfrak{S}}}=0,$$

and we have therefore

$$-\sqrt{\mathfrak{S}} = \sqrt{X}\,\frac{dC}{dx} = \sqrt{Y}\,\frac{dC}{dy};$$

viz. $\sqrt{X}\,\dfrac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before

$$M = \frac{\sqrt{X}-\sqrt{Y}}{x-y},$$

we have

$$C = M^2-d(x+y)-e(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\,\frac{dM}{dx}-d-2e(x+y).$$

We have

$$\frac{dM}{dx} = \frac{1}{x-y}\,\frac{X'}{2\sqrt{X}}-\frac{\sqrt{X}-\sqrt{Y}}{(x-y)^2},$$

and hence

$$\begin{aligned} \sqrt{\mathfrak{S}}\,(x-y)^2 &= \sqrt{X}\,(x-y)^2\Big\{2M\,\frac{X'}{2\sqrt{X}}(x-y)-d-2e(x+y)\Big\} \\ &= -(x-y)X'(\sqrt{X}-\sqrt{Y})+2(X+Y-2\sqrt{XY})\sqrt{X} \\ &\quad+(d+2e(x+y))(x-y)^2\sqrt{X} \\ &= \big[(x-y)X'+2X+2Y+(d+2e(x+y))(x-y)^2\big]\sqrt{X} \\ &\quad+\big[(x-y)X'-4X\big]\sqrt{Y}. \end{aligned}$$

We obtain at once the coefficient of $\sqrt{Y}$, and with little more difficulty that of $\sqrt{X}$; and the result is

$$\begin{aligned} \sqrt{\mathfrak{S}}\,(x-y)^2 = &-\big[4a+3bx+2cx^2+dx^3+y(b+2cx+3dx^2+4ex^3)\big]\sqrt{Y} \\ &+\big[4a+3by+2cy^2+dy^3+x(b+2cy+3dy^2+4ey^3)\big]\sqrt{X}. \end{aligned}$$

We have also

$$\begin{aligned} C(x-y)^2 &= (\sqrt{X}-\sqrt{Y})^2 - d(x+y)(x-y)^2 - e(x+y)^2(x-y)^2 \\ &= X+Y-d(x^3-x^2y-xy^2+y^3)-e(x^4-2x^2y^2+y^4)-2\sqrt{XY} \\ &= 2a+b(x+y)+c(x^2+y^2)+dxy(x+y)+2ex^2y^2-2\sqrt{XY}, \end{aligned}$$

or, say

$$\begin{aligned} C(x-y)^2 = 2a(x-y)+b(x^2-y^2)+c(x^3-x^2y+xy^2-y^3)& \\ +dxy(x^2-y^2)+2ex^2y^2(x-y)-2(x-y)\sqrt{XY}.& \end{aligned}$$

We can hence form the expression of

$$(x-y)^2\Big\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\Big\},$$

viz. this is

$$\begin{aligned} &= (2be-cd)(x-y)^2+2ad(x-y)+bd(x^2-y^2)+cd(x^3-x^2y+xy^2-y^3) \\ &\quad + d^2xy(x^2-y^2)+2de\,x^2y^2(x-y)-2d(x-y)\sqrt{XY} \\ &\quad + 2\sqrt{e}\{[-(4a+3bx+2cx^2+dx^3)-y(b+2cx+3dx^2+4ex^3)]\sqrt{Y} \\ &\qquad + [(4a+3by+2cy^2+dy^3)+x(b+2cy+3dy^2+4ey^3)]\sqrt{X}\}, \end{aligned}$$

and this should be

$$\begin{aligned} &= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2) \\ &\qquad + 2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})-2\sqrt{XY}\big\} \\ &\quad \times \big\{d(x-y)+2e(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\big\}. \end{aligned}$$

The function on the right-hand is, in fact,

$$\begin{aligned} &= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)-2\sqrt{XY}\big\} \\ &\quad \times \big[d(x-y)+2e(x^2-y^2)\big]+4e(x-y)(\sqrt{X}-\sqrt{Y})(x\sqrt{Y}-y\sqrt{X}) \\ &\quad + 2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)-2\sqrt{XY}\big\} \\ &\quad + 2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})\big[d(x-y)+2e(x^2-y^2)\big], \end{aligned}$$

viz. this is

$$\begin{aligned} &= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\big\} \\ &\quad \times \big\{d(x-y)+2e(x^2-y^2)\big\}+4e(x-y)(-xY-yX) \\ &\quad - 2\sqrt{XY}\big[d(x-y)+2e(x^2-y^2)\big]+4e(x-y)(x+y)\sqrt{XY} \\ &\quad + 2\sqrt{e}\,\Big(\sqrt{X}\big[2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2) \\ &\qquad\qquad + 2Y-(x-y)y\big(d(x-y)+2e(x^2-y^2)\big)\big] \\ &\qquad - \sqrt{Y}\big[2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2) \\ &\qquad\qquad + 2X-(x-y)x\big(d(x-y)+2e(x^2-y^2)\big)\big]\Big), \end{aligned}$$

which is, in fact, equal to the expression on the left-hand side.

To complete the theory, we require to express $\sqrt{Z}$ as a function of x, y. It would be impracticable to effect this by direct substitution of the foregoing value of z; but, observing that the value in question is a solution of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}+\dfrac{dz}{\sqrt{Z}}=0$, or, what is the same thing, that $\dfrac{1}{\sqrt{X}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dx}=0$, $\dfrac{1}{\sqrt{Y}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dy}=0$, we may from either of these equations, considering therein z as a given function of x, y, calculate $\sqrt{Z}$.

Writing for shortness

$$z = \frac{J - 2\sqrt{e}\,y(x-y)\sqrt{X} + 2\sqrt{e}\,x(x-y)\sqrt{Y} - 2\sqrt{XY}}{R - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}},$$

where

$$R = (x-y)^2[d + 2e(x+y)]\,,$$
$$J = 2a + b(x+y) + 2cxy + dxy(x+y) + 2exy(x^2 - xy + y^2);$$

or, if for a moment $z = \dfrac{N}{D}$, then

$$\frac{dz}{dx} = \frac{1}{D^2}\left(D\,\frac{dN}{dx} - N\,\frac{dD}{dx}\right) = -\frac{\sqrt{Z}}{\sqrt{X}},$$

that is,

$$\sqrt{Z} = \frac{\sqrt{X}}{D^2}\left(N\,\frac{dD}{dx} - D\,\frac{dN}{dx}\right) = \frac{\Omega}{D^2}\ \text{suppose;}$$

or, writing for shortness X', R', J' to denote the derived functions $\dfrac{dX}{dx}$, $\dfrac{dR}{dx}$, $\dfrac{dJ}{dx}$ (Y' is afterwards written to denote $\dfrac{dY}{dy}$, but as the final formulae contain only $X' = \dfrac{dX}{dx}$, and $Y' = \dfrac{dY}{dy}$, this does not occasion any defect of symmetry), we find

$$\begin{aligned}\Omega = {} & N[R'\sqrt{X} - 2\sqrt{e}\,X - \sqrt{e}\,(x-y)X' + 2\sqrt{e}\,\sqrt{XY}] \\ & - D[J'\sqrt{X} - 2\sqrt{e}\,yX - \sqrt{e}\,(x-y)yX' + 2\sqrt{e}\,(2x-y)\sqrt{XY} - X'\sqrt{Y}];\end{aligned}$$

and substituting herein for N, D their values, and arranging the terms, we find

$$\Omega = \mathfrak{A} + \mathfrak{B}\sqrt{X} + \mathfrak{C}\sqrt{Y} + \mathfrak{D}\sqrt{XY},$$

where

$$\begin{aligned}\mathfrak{A} = {} & -J\,[2X + (x-y)X'] \\ & - 2(x-y)yR'X \\ & + Ry\{2X + (x-y)X'\} \\ & + 2(x-y)XJ' \\ & + 2(x-y)X'Y,\end{aligned} \qquad \begin{aligned}\mathfrak{B} = {} & JR' \\ & + 2e(x-y)y\{2X + (x-y)X'\} \\ & + 4ex(x-y)Y \\ & - RJ' \\ & - 2e(x-y)y[2X + (x-y)X'] \\ & - 4e(x-y)(2x-y)Y,\end{aligned}$$

$$\begin{aligned}\mathfrak{C} = {} & -4ey(x-y)X \\ & - 2e(x-y)x[2X + (x-y)X'] \\ & - 2R'X \\ & + RX' \\ & + 2e(x-y)y\{2X + (x-y)X'\} \\ & + 4e(x-y)(2x-y)X,\end{aligned} \qquad \begin{aligned}\mathfrak{D} = {} & 2J \\ & + 2(x-y)xR' \\ & + 2[2X + (x-y)X'] \\ & - 2(2x-y)R \\ & - 2(x-y)X' \\ & - 2(x-y)J,\end{aligned}$$

where the terms have been written down as they immediately present themselves; but, collecting and arranging, we have

$$\begin{aligned}\mathfrak{A} &= 2X(-J + Ry - 2Y) + (x-y)[2XJ' + 2X'Y - X'J - 2yR'X + yRX'], \\ \mathfrak{B} &= JR' - J'R - 4e(x-y)^2Y, \\ \mathfrak{C} &= -4XR' + X'R + 4e(x-y)^2X - 2e(x-y)^3X', \\ \mathfrak{D} &= 2J + 4X - 2Rx + 2(x-y)(xR' - R - J').\end{aligned}$$

To reduce these expressions, writing

$$M = d + 2e(x+y),$$

$$\Lambda = c + d(x+y) + e(x^2+y^2),$$

we have $R = (x-y)^2M$, and therefore $R' = 2(x-y)M + 2e(x-y)^2$; also

$$J = X + Y - (x-y)^2\Lambda;$$

also, from the original form,

$$J' = b + 2cy + d(2xy+y^2) + e(6x^2y - 4xy^2 + 2y^3).$$

The final values are

$$\begin{aligned}
\mathfrak{A} &= -X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b+dxy)M + xyM^2\},\\
\mathfrak{B} &= (x-y)M[4Y + (x-y)Y'] + 2e(x-y)^3Y',\\
\mathfrak{C} &= -(x-y)M[4X - (x-y)X'] - 2e(x-y)^3X',\\
\mathfrak{D} &= 4(X+Y) + 4e(x-y)^4,
\end{aligned}$$

which, once obtained, may be verified without difficulty.

Verification of $\mathfrak{A}$.—The equation is

$$-X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b+dxy)M + xyM^2\}$$

$$= 2X(-J + Ry - 2Y) + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\};$$

or, putting for shortness

$$\Lambda^2 + (-b+dxy)M + xyM^2 = \nabla,$$

this is

$$\begin{aligned}
(x-y)^4\nabla &= X^2 + 6XY + Y^2\\
&\quad + 2X\{-X - 3Y + (x-y)^2\Lambda + (x-y)^3yM\}\\
&\quad + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\}\\
&= -X^2 + Y^2 + 2(x-y)^2X\Lambda + 2(x-y)^3yXM\\
&\quad + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\};
\end{aligned}$$

we have $-X^2 + Y^2 = -(X-Y)(X+Y)$, where $X - Y$ divides by $x - y$, $= (x-y)\Pi$ suppose; hence, throwing out the factor $x - y$, the equation becomes

$$\begin{aligned}
(x-y)^3\nabla &= -\Pi(X+Y) + 2(x-y)X\Lambda + 2(x-y)yXM\\
&\quad + 2XJ' + 2X'Y - X'\{X + Y - (x-y)^2\Lambda\}\\
&\quad - 2yX\left[2(x-y)M + 2(x-y)^2e\right] + (x-y)^2yMX'\\
&= -\Pi(X+Y) + 2XJ' - X'(X-Y)\\
&\quad + 2(x-y)X\Lambda - 2(x-y)yXM\\
&\quad + (x-y)^2X'\Lambda - 4(x-y)^2eyX + (x-y)^2yMX'.
\end{aligned}$$

We have $2XJ' = J'(X+Y) + J'(X-Y)$, and hence the first line is

$$= (-\Pi + J')(X+Y) + J'(X-Y);$$

$-\Pi + J'$, as will be shown, divides by $x - y$, or say it is $= (x-y)\Phi$, and, as before, $X - Y$ is $= (x-y)\Pi$; hence, throwing out the factor $x - y$, the equation becomes

$$(x-y)^2\nabla = \Phi(X+Y) + \Pi(J' - X') + 2X\Lambda - 2yXM + (x-y)[X'\Lambda - 4eyX + yMX'].$$

We have

$$\Pi = b + c(x+y) + d(x^2+xy+y^2) + e(x^3+x^2y+xy^2+y^3),$$

and thence

$$-\Pi + J' = c(-x+y) + d(-x^2+y^2) + e(-x^3+3x^2y-3xy^2+y^3);$$

or, dividing this by $(x-y)$, we find

$$\Phi = -c - dx - e(x^2-xy+y^2),$$

or, as this may be written,

$$\Phi = -\Lambda + dy + 2exy.$$

We find, moreover,

$$J' - X' = 2c(-x+y) + d(-3x^2+2xy+y^2) + e(-4x^3+6x^2y-4xy^2+2y^3),$$

which divides by $(x-y)$, the quotient being

$$-2c - d(3x+y) - e(4x^2-2xy+2y^2),$$

viz. this is

$$= -2\Lambda - (x-y)(d+2ex).$$

Hence the equation now is

$$\begin{aligned}(x-y)^2\nabla = (X+Y)[-\Lambda + dy + 2exy] + 2X\Lambda - 2yXM \\ + (x-y)\Pi\{-2\Lambda - (x-y)(d+2ex)\} \\ + (x-y)\{X'\Lambda - 4eyX + yMX'\}.\end{aligned}$$

The first line is

$$(X+Y)[-\Lambda + yM + 2(x-y)ye] + 2X\Lambda - 2yXM,$$

which is

$$= (\Lambda - yM)(X-Y) + 2(x-y)ey(X+Y);$$

hence, throwing out the factor $x-y$, the equation becomes

$$\begin{aligned}(x-y)\nabla &= (\Lambda - yM)\Pi + 2ey(X+Y) - 2\Lambda\Pi + X'\Lambda - 4eyX + yMX' - (x-y)\Pi(d+2ex) \\ &= (\Lambda + yM)(-\Pi + X') - 2ey(X-Y) - (x-y)\Pi(d+2ex).\end{aligned}$$

We have

$$-\Pi + X' = c(x-y) + d(2x^2-xy-y^2) + e(3x^3-x^2y-xy^2-y^3),$$

which is $= (x-y)(\Lambda + xM)$: also $(X-Y) = (x-y)\Pi$, as before; whence, throwing out the factor $x-y$, the equation is

$$\nabla = (\Lambda + xM)(\Lambda + yM) - 2ey\Pi - (d+2ex)\Pi,$$

that is,

$$\nabla = (\Lambda + xM)(\Lambda + yM) - M\Pi,$$

viz. substituting for ∇ its value, reducing, and throwing out the factor M, the equation becomes

$$-b + dxy = (x+y)\Lambda - \Pi,$$

which is right.

Verification of $\mathfrak{B}$.—The equation is

$$J[2(x-y)M + 2e(x-y)^2] - J'(x-y)^2M - 4(x-y)^2Y$$

$$= 4(x-y)MY + (x-y)^2MY' + 2e(x-y)^3Y',$$

which, throwing out the factor $x-y$, is

$$0 = 2M(-J+2Y) + (x-y)M(J'+Y') + 2e(x-y)(-J+2Y) + 2e(x-y)^2Y'.$$

Here $-J+2Y$, $= -(X-Y)+(x-y)^2\Lambda$, is divisible by $(x-y)$: hence, throwing out the factor $x-y$, the equation is

$$\begin{aligned}0 = {} & M\{-2b-2c(x+y)-2d(x^2+xy+y^2)-2e(x^3+x^2y+xy^2+y^3)\} \\ & + M(J'+Y') + 2M(x-y)\Lambda + 2e(-J+2Y) + 2e(x-y)Y'.\end{aligned}$$

In the first and second terms, the factor which multiplies M is

$$c(-2x+2y) + d(-2x^2+2y^2) + e(-2x^3+2x^2y-6xy^2+6y^3),$$

which is divisible by $x-y$; also $-J+2Y$, $= -(X-Y)+(x-y)^2\Lambda$, is divisible by $(x-y)$: hence, throwing this factor out, the equation is

$$\begin{aligned}0 = {} & M\{-2c+d(-2x-2y)+e(-2x^2+2xy-4y^2)\} + 2M\Lambda \\ & + 2e\{-b-c(x+y)-d(x^2+xy+y^2)-e(x^3+x^2y+xy^2+y^3)\} \\ & + 2e(x-y)\Lambda + 2eY'.\end{aligned}$$

Here in the first line the coefficient of M is $= e(2xy-2y^2)$: hence, throwing out the constant factor $2e$, the equation is

$$0 = -b-c(x+y)-d(x^2+xy+y^2)-e(x^3+x^2y+xy^2+y^3)+Y'+(x-y)yM+(x-y)\Lambda.$$

The first five terms are

$$= c(-x+y)+d(-x^2-xy+2y^2)+e(-x^3-x^2y-xy^2+3y^3),$$

which is divisible by $x-y$; throwing out this factor, the equation is

$$0 = -c-d(x+2y)-e(x^2+2xy+3y^2)+\Lambda+yM,$$

which is right.

Verification of $\mathfrak{C}$.—We have

$$\begin{aligned} & -2X[2(x-y)M+2e(x-y)^2]+(x-y)^2X'M+4e(x-y)^2X-2e(x-y)^3X' \\ & = -(x-y)M[4X-(x-y)X']-2e(x-y)^3X'\end{aligned}$$

which is, in fact, an identity.

Verification of $\mathfrak{D}$.—The equation may be written

$$\begin{aligned}4X+4Y+4e(x-y)^4 = {} & 2X+2Y-2(x-y)^2\Lambda \\ & +4X-2x(x-y)^2M \\ & +2(x-y)\{2(x-y)xM+2ex(x-y)^2-M(x-y)^2-J'\},\end{aligned}$$

viz. this is

$$= 2X-2Y-4e(x-y)^4-2(x-y)^2\Lambda+2x(x-y)^2M+4ex(x-y)^3-2M(x-y)^3-2(x-y)J'.$$

The first term $2(X-Y)$ is divisible by $2(x-y)$; throwing this factor out, the equation becomes

$$\begin{aligned} & = b+c(x+y)+d(x^2+xy+y^2)+e(x^3+x^2y+xy^2+y^3)-J' \\ & \quad -2e(x-y)^3-(x-y)\Lambda+x(x-y)M+2ex(x-y)^2-M(x-y)^2.\end{aligned}$$

Substituting for J' its value, the first line becomes

$$c(x-y)+d(x^3-xy^2)+e(x^4-5x^2y+5xy^2-y^4),$$

which is divisible by $(x-y)$; hence, throwing out this factor, the equation is

$$= c + dx + e(x^2 - 4xy + y^2) - \Lambda + xM - 2e(x-y)^2 + 2ex(x-y) - M(x-y),$$

where the sum of all the terms but the last is $= d(x-y) + e(2x^2 - 2xy)$: hence, again throwing out the factor $x - y$, the equation becomes

$$= d + 2ex - 2e(x-y) + 2ex - M,$$

which is right.

Recapitulating, we have for the general integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, or the particular integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} + \dfrac{dz}{\sqrt{Z}} = 0$,

$$z = \frac{J - 2\sqrt{e}\,y(x-y)\sqrt{X} + 2\sqrt{e}\,x(x-y)\sqrt{Y} - 2\sqrt{XY}}{(x-y)^2M - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}},$$

the corresponding value of $\sqrt{Z}$ being

$$\begin{aligned}\sqrt{Z} = \Big\{\sqrt{e}\big[-X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b + dxy)M + xyM^2\}\big]&\\ + \big[\{4Y + (x-y)Y'\}M + 2e(x-y)^2Y'\big](x-y)\sqrt{X}&\\ - \big[\{4X - (x-y)X'\}M + 2e(x-y)^2X'\big](x-y)\sqrt{Y}&\\ + \big[4(X+Y) + 4e(x-y)^4\big]\sqrt{XY}\Big\}&\\ \Big/\big\{(x-y)^2M - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}\big\}^2&,\end{aligned}$$

where, as before,

$$M = d + 2e(x+y),$$
$$\Lambda = c + d(x+y) + e(x^2+y^2),$$
$$J = 2a + b(x+y) + 2cxy + dxy(x+y) + 2exy(x^2 - xy + y^2);$$

also X is the general quartic function $a + bx + cx^2 + dx^3 + ex^4$, and Y, Z are the same functions of y, z respectively.

In connexion with what precedes, I give some investigations relating to the more simple form $\Theta = a + c\theta^2 + e\theta^4$, or, as it will be convenient to write it, $\Theta = 1 - l\theta^2 + \theta^4$.

We have

$$\begin{vmatrix} x, & \sqrt{X} \\ y, & \sqrt{Y} \end{vmatrix} = 0 \quad \text{a particular integral} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0;$$

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \end{vmatrix} = 0 \quad \begin{matrix}\text{the general integral}\\ \text{a particular integral}\end{matrix} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0;$$

$$\begin{vmatrix} x^3, & x, & x^2\sqrt{X}, & \sqrt{X} \\ y^3, & y, & y^2\sqrt{Y}, & \sqrt{Y} \\ z^3, & z, & z^2\sqrt{Z}, & \sqrt{Z} \end{vmatrix} = 0 \quad \begin{matrix}\text{the general integral}\\ \text{a particular integral}\end{matrix} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0;$$

$$\begin{vmatrix} x^3, & x, & x^2\sqrt{X}, & \sqrt{X} \\ y^3, & y, & y^2\sqrt{Y}, & \sqrt{Y} \\ z^3, & z, & z^2\sqrt{Z}, & \sqrt{Z} \\ w^3, & w, & w^2\sqrt{W}, & \sqrt{W} \end{vmatrix} = 0 \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} = 0,$$

and so on; viz. in taking

$$\begin{vmatrix} x^2, & x, & \sqrt{X} \\ y^2, & y, & \sqrt{Y} \\ z^2, & z, & \sqrt{Z} \end{vmatrix} = 0 \quad \text{as the general integral of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

we consider z as the constant of integration: and so in other cases.

It is to be remarked that it is an essentially different problem to verify a particular integral and to verify a general integral, and that the former is the more difficult one. In fact, if $U = 0$ is a particular integral of the differential equation $M\,dx + N\,dy = 0$, then we must have $N\,\dfrac{dU}{dx} - M\,\dfrac{dU}{dy} = 0$, not identically but in virtue of the relation $U = 0$, or we have to consider whether two given relations between x and y are in fact one and the same relation. In the case of a general solution, this is theoretically reducible to the form $c = V$, c being the constant of integration, and we have then the equation $N\,\dfrac{dV}{dx} - M\,\dfrac{dV}{dy} = 0$, satisfied identically, or, what is the same thing, U a solution of this partial differential equation.

Hence it is theoretically easier to verify that

$$\begin{vmatrix} x^2, & x, & \sqrt{X} \\ y^2, & y, & \sqrt{Y} \\ z^2, & z, & \sqrt{Z} \end{vmatrix} = 0$$

is a general solution, than to verify that

$$\begin{vmatrix} x, & \sqrt{X} \\ y, & \sqrt{Y} \end{vmatrix} = 0$$